19 At the start of year n, the number of animals in a population is P_n

At the start of the following year, the number of animals in the population is P_{n+1} where

$$P_{n+1} = kP_n$$

At the start of 2017 the number of animals in the population was 4000
At the start of 2019 the number of animals in the population was 3610

Find the value of the constant k.

Two annual changes: $3610 = 4000k^2$

$k^2 = 3610/4000 = 0.9025$

$k = \sqrt{0.9025} = 0.95$

1 mark
apply annual factor twice

1 mark
solve square equation

1 mark
k=0.95

0.95

(Total for Question 19 is 3 marks)

20 Pat throws a fair coin n times.

Find an expression, in terms of n, for the probability that Pat gets at least 1 head and at least 1 tail.

(Total for Question 20 is 2 marks)

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